A Fuzzy Logic Based Frugality Evaluation Score For Machine Learning Models

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***Abstract*—The use of artificial intelligence (AI) has risen exponentially in the last decade, and the popularity of frugal AI followed this trend in the community. Yet, most of this research focuses on efficiency as in decreasing the energy consumption and raising the performance. Frugality is a term that accounts for the objectives of an application, and encapsulates their subjective aspect. Indeed, what is a good performance or a low energy consumption? These terms are difficult to quantify and the question is task-specific. In this work, we propose a new frugality scoring method based upon fuzzy logic to address this issue. The proposed frugality scoring method is applied to a number of common tasks in machine learning, which provides a wide evaluation of the plasticity of the fuzzy based scoring. This score allows for an objective-aware evaluation of the frugality of a method depending on the user's criteria.**

***Index Terms*—Frugality, machine learning, scoring, fuzzy logic**

I. Introduction

The past decade has seen an exponential increase in energy consumption across different domains of computer science, especially with the rise of artificial intelligence (AI) usage by the general public. Recent projections show an increase of the global electricity usage around 210.6 TWh per year from 2027 [1] with the latest trends in GPU manufacturing and generative AI usage. In a high-efficiency scenario where progress in energy consumption savings follows the rising demand, the International Energy Agency predicts an annual electricity demand for AI of 970 TWh by 2035 [2]. On the other hand, cases of AI development favouring transitions to greener electricity production systems have been highlighted when coordinated governance policies included focuses on both sustainability and performance [3]. The term **performance** is used to qualify the level of adequacy of the results provided by the method for a designated task relative to an objective. It is evaluated through different metrics such as accuracy, error rate, precision or recall for classification tasks [4], or cluster purity, rand index or entropy for clustering tasks [5] for example. Therefore, the term performance here focuses only on the results provided by the application of a method, regardless of its implementation. Facing these stakes, the machine learning community aims for more efficiency.

A. From Efficiency To Frugality

There is a broad focus on efficiency within the machine learning (ML) literature [6], which aims for an optimization of both the performance and the energy costs. To evaluate how efficient an application is, a growing practice is to measure its energy consumption, for example the energy consumption during the training phase of a neural network, and to put it in relation with its performance (its accuracy, for instance). Researchers show that their model has a higher accuracy than other state-of-the-art models, and a lower estimated energy consumption or less runtime during training. In rare (but more and more frequent) cases, they translate this energy consumption into carbon emissions. Some papers will compare these emissions to reference quantities making sense to users, such as flight travel emissions or average household electricity consumptions [7]. Identifying whether these costs are elevated or not is also a difficult task if we only focus on the scalar values. Thus, a metric constructed to be easily interpretable for the user can simplify the evaluation, but every metric makes sense if we consider it in the context of its applications. Indeed, the expected energy consumptions and performance for a deep learning model could not be similar to the ones for a more simple non-neural ML method. This is why the term **frugality** emerges, as a paradigm that aims for a development and usage that meet the actual reality of the needs for an application [8]. The type of application considered puts the studied task in a context, and the objectives in terms of energy consumption and performance are then defined in this context.

B. Scoring Approaches And Their Limits

Two strategies can be proposed in order to evaluate the frugality of a method. One is to consider each dimension (the performance or the energy costs) independently, making a decision based on the trends observed on each dimension. This has proven to be complex for systems such as those where some dimensions tend to increase while others decrease [9]. The other approach is to aggregate these dimensions as to create a new mono-dimensional score single-handedly capturing the frugality of a method. This approach is akin to consumer advice mechanisms, like scores providing insight on the nutritional values of food products in distribution stores, and such scores are currently in use. France adopted such the Nutri-score [10] in 2016 to invite consumers to turn to healthier food habits. This approach was then adopted in Belgium, Spain, Switzerland and Luxemburg after studies showed a positive impact on consumer behaviours toward more nutritional products [11]. The Eco-score, evaluating food products' ecological impacts, was also adopted with similar results on consumer behaviour regarding ecological impact. Although, this behaviour was less verified for products showing contradictory scores, like products showing a positive Nutri-score, and a negative Eco-score [12]. This is the main issue, how to combine information into a single metric when there are multiple objectives, and when the choice is inherently human? The literature provides several scores to aggregate the performance variable α and energy consumption β. A direct weighted sum S_{WS} [13] presented in equation (1) or a combination using a weighted harmonic mean S_{HM} [14] in equation (2) provide simple weighted aggregations:

$$S_{\text{WS}}(\alpha, \beta, \varepsilon, B) = \varepsilon \times \alpha + (1-\varepsilon) \times (1 - \frac{\beta}{B}), \quad (1)$$

$$S_{\text{HM}}(\alpha, \beta, \kappa, B) = (1+\kappa^2)\frac{\alpha(1-\frac{\beta}{B})}{\kappa^2\alpha + (1-\frac{\beta}{B})}. \quad (2)$$

A weight $\varepsilon \in [0, 1]$ and $\kappa \in \mathbb{R}^+$ is given to the energy consumption variable over the performance in S_{WS} and S_{HM} respectively, and necessitate the information of maximum energy consumption reached B for normalisation. But the choice of weight is poorly interpretable, and the maximum energy consumption reached B is hard to predict in advance. Evchenko et al (2021) [15] provide a efficiency score S_{E} in equation (3) built on the non-normalized value of β, and with a similar weight parameter $\text{w} \in \mathbb{R}^+$:

$$S_{\text{E}}(\alpha, \beta, \text{w}) = \alpha - \frac{\text{w}}{1+\frac{1}{\beta}}. \quad (3)$$

Yet, these three score only provide a evaluation of the trade-off between the performance and energy consumption. A frugality evaluation metric requires context-dependency and interpretability to not focus solely on efficiency. Fuzzy logic stands out as a relevant domain to consider these requirements.

C. Proposed Approach

Dealing with subjective data interpretation is a standard task in the fuzzy logic domain. Yet, no current research focuses on the consideration of the frugality of an AI application as a fuzzy-based control system as for the present authors knowledge. In this work, we explore the potential of this consideration that can encapsulate the subjectivity and task dependency of frugality. Here are the main contributions of this paper:

- This work provides a frugality score that is interpretable, bounded and parametrizable to user-defined objectives. It allows for an evaluation of a candidate method based on both its performance and its energy consumption, and can be parametrized to include different types of energy consumptions. We compare its relevance with other scores from the literature to highlight how it best encapsulate the definition of frugality.
- We propose a validation of our evaluation framework over multiple classical ML scenarios, as to highlight the context-dependency of frugality in ML.
- We also provide a ready-for-use package[1] for Python users, and a direct visualisation dashboard[2] for this evaluation framework.

[1] https://anonymous.4open.science/r/frugscore/
[2] https://anonymous.4open.science/r/frugscore/doc/webapp/index.html

D. Paper Structure

The rest of this paper is structured as follow: Section II presents the related works in Green AI, efficiency scoring and fuzzy logic. In Section III, the proposed frugality score framework is formally introduced. Section IV presents the experimental setup before Section V covers the results. Finally, Section VI and VII respectively present the discussions about this work and the conclusions and future directions.

II. Related Work

A. Green AI and Energy Measurements

The past decade, research for energy efficiency and sustainability in AI has been rising as well as the ecological footprint of the AI industry. The term Green AI became a specific field in research to define AI applications in which sustainability is taken into account during development, as opposed to Red AI for which accuracy is the main incentive at the cost of the carbon footprint [6]. The term *Green-in AI* is also used to define these aspects to separate such applications from AI tasks which aim for a positive impact on the ecological crises. These are labelled under *Green-by AI*. Different strategies under the label *Green-in AI* are present in the literature, depending on the focus within the data processing chain. While some works apply selective sampling strategies for the training data to reduce the training duration and energy consumption [16], using specific hardware and parallelization strategies are commonly used to speed up training [6]. Classical strategies for neural networks also include specific architectures with pruning [17] and quantization [18], and implementations of distributed learning [19]. These methods usually aim for the reduction of both the runtime and the energy consumption during training. Yet, if measuring the first metric is straightforward, assessing the energy consumption of a ML application an experiment [20] is a complex task.

Measuring the energy consumption empirically allows for an accurate description that does not involve having prerequisite about the exact energy consuming processes. It is relatively simple to apply, but faces many challenges. The measurement error is tightly linked to the precision of the sensor, and the description of the energy consumption is only global. Connecting the sensor to specific part of the system can be invasive. The frequency of the measurement is usually low in sensors that are affordable for the general public, which makes them more adapted to applications with extensive running time. Approximating the energy consumption through other metrics is also a classical approach within the green AI community. Thus, several energy consumption evaluation softwares have been published, such as CodeCarbon [21], CarbonTracker [22] or Experiment Impact Tracker [23] for Python, each adapted to different applications [24]. This approach addresses the invasive aspect of a specific energy consumption sensor and is more easily reproducible. It can also provide a higher tracking frequency given model parameters with higher frequencies (such as CPU or GPU usage). However, as an approximation, this method can be biased by the model used to evaluate the energy consumption, and can miss hidden costs or abnormalities which could transpire in real time energy consumption measurements. It is only once this dimension is quantified that one can evaluate the efficiency of their ML method.

B. Efficiency and Frugality scoring

As research in Green AI rose exponentially, the interest in an evaluation metric to quantify the effectiveness of the effort on sustainability followed. Efficiency scoring methods aim for the identification of metrics that are correlated positively to the performance, and negatively to the energy consumption. They include theoretical approaches based on the accuracy over number of parameters of a model [25], [26], while empirical approaches focus on runtime and power measurements to relatively compare Green IA methods [15]. However, some academics debate that running time is sufficient to represent energy consumption [27]. A recent work that related to this paper was Yang and Armour's finding about the energy consumption during inference for deep learning models [13]. The authors' work on the combination of energy consumption and accuracy in a distance metric to an ideal zero-consuming all-solving computer vision model allows a direct evaluation of the energy efficiency of a model. However, evaluating the frugality of a method differs from these approaches by integrating pre-chosen evaluated objectives, which introduce human subjectivity.

The term frugality is discussed in the literature and is associated to several key concepts. Frugal innovation is classically defined under the following qualities: functional, robust, user-friendly, growing, affordable, and local [28]. These concepts also stand out in machine learning in with its own definition of frugality, and frugal aims for systems that are efficient, lightweight, and resource-constrained [29]. In this sense, efficiency is not far from frugality, as both aim for a performance conservation or improvement with minimal resources, such as electric power, memory, water, or training data. Yet, other work find a more specific definition for frugality rooted in sustainable innovation [8] and sufficiency in the definition set by the IPCC [30]. According to this view, frugality aims for a redefinition of the objectives of a method rather than a direct optimization. Identifying the need goes in pair with the resource usage. Put simply, instead of improving the performance while minimizing the costs, frugality focuses on defining what performance is acceptable and what resource is available before combining both information to make a decision. However, such statements are inherently subjective and context-dependant, and require human-in-the-loop intervention. Fuzzy logic provides a solution to these criteria.

C. Fuzzy logic for multi-criteria evaluation

Fuzzy systems are especially relevant for applications in which the appreciation of a quality for a certain quantitative measure is vague or subject to human interpretation. It is commonly used in multi-criteria decision-making [31], [32] and also risk assessment such as for natural hazard risk evaluation [33], [34]. What makes fuzzy logic adapted for these tasks is that it is specifically viable when dealing with problems that are unstructured and intuitive [35] and it allows to incorporate expert knowledge within its model. Therefore, applying this type of tool for sustainability problems is relevant in the sense that the appreciation of the performance of a method is often subjective and prone to be a human decision. It is then why several fuzzy logic models have been published to assess sustainability within the research community on management practices and policies. The model SAFE [36], generally accepted for project sustainability assessment, aims to obtain three sustainability assessment measures (ecological system integrity, human system integrity and overall sustainability) from a range of primary sustainability indicators, such as CO_2 emissions, water pollution or energy production. Fuzzy systems are also generally considered for large scale systems such as countries sustainability assessment [37]. Here, we propose an application of this paradigm on a system of a smaller scale, the application of a machine learning model. This model is presented in section III.

III. Proposed Framework

A. Problem Formulation

In this work, we consider a frugality score that takes as input variables a performance α_i and an energy consumption β_i for an execution i. Let be S our score, O a set of objectives for the input variables and M a set of machine learning methods which are subject to evaluation. A frugality score $S(\alpha_i, \beta_i, O(M))$ is context-dependent and relies on a set of objectives $O(M)$ specific to the candidate methods. When O is independent from the application context, the score $S(\alpha_i, \beta_i, O)$ focuses solely on efficiency. Furthermore, we aim for a frugality score to follow a monotonicity rule regarding the performance and energy consumptions, so $S(\alpha_i, \beta_i, O(M))$ must satisfy $\frac{\partial S}{\partial \alpha_i} \geq 0$ and $\frac{\partial S}{\partial \beta_i} \leq 0$ for all experiment i.

B. Fuzzy Inference System

In fuzzy logic, we define fuzzy sets as keywords with a degree of truth between 0 and 1, and an associated membership function. We define a membership function μ_{D_X} as in equation (4), which associates the value of a variable X in its universe of discourse Ω_X and the descriptive keyword D_X of its fuzzy set with a degree of truth in $[0, 1]$:

$$\mu_{D_X} : \begin{cases} \Omega_X \longrightarrow [0,1] \\ x \longmapsto \max\left(\min\left(\dfrac{x-a}{b-a}, \dfrac{d-x}{d-c}, 1\right), 0\right) \end{cases}$$
$$\text{with } (a,b,c,d) \in (\Omega_X)^4. \tag{4}$$

In this work, we model information fusion with a Mamdani Fuzzy Inference System (FIS) [38], for which the inference process is summarized in Algorithm 1. For each experiment i, we fuzzify the energy consumptions and performance metrics, fire the association rules, and defuzzify the aggregated the output. These step are detailed in the following sections.

C. Membership functions

The core of the Mamdani fuzzy inference system resides in the definition of membership functions which map each point

Algorithm 1 Mamdani Inference System

Input: energy consumption $\beta_i \in \Omega_\beta$, accuracy $\alpha_i \in \Omega_\alpha$ for an experiment i
for $j = 1$ **to** 5 **do**
  Initialize $L_j = \emptyset$, $\Sigma_j = \emptyset$.
  for $(D_\beta,\ D_\alpha)$ **in** rule matrix **where** $D_S = j$ **do**
    L_j.append$\big(min\big(\mu_{D_\beta}(\beta_i), \mu_{D_\alpha}(\alpha_i)\big)\big)$
  end for
  for $s \in \Omega_S$ **do**
    Σ_j.append$\big(min\big(max(L_j), \mu_j(s)\big)\big)$
  end for
end for
Output: $\Phi\big(\bigcap_{j=1}^{5}(\Sigma_j)\big)$

within the universe of discourse of a variable to a membership value within $[0, 1]$ associated to a descriptive keyword. In our case, we consider the energy consumption of an algorithm in joules, and its universe of discourse $\mathbb{R}^+$, which we want to associate to the descriptive keyword ”high”, ”medium” and ”low”:

- low energy : the execution time fits within the expected operational duration,
- medium energy : the execution time remains acceptable,
- high energy : the execution time exceeds practical expectations.

Thus these keywords, labelled D_β, are associated with their respective membership functions in Fig. 1. For interpretability, we choose to fix the expected energy consumptions based on the expected execution time of the application, here under 1 day, between 1 and 3 days, and more than 3 days. The chosen energy consumptions thresholds are then 2.38×10^7 J and 7.13×10^7 J, corresponding to the thresholds delimiting the three ranges of energy consumption for a 275 W system. All chosen parameters of this work are summarized in Table III. Membership functions are also chosen for the accuracy of an algorithm, defining three keywords D_α to qualify its performance (”low”, ”medium” and ”high”). Similarly to the energy consumption,

- high performance : the performance metric fits within the objective,
- medium performance : the performance metric remains acceptable,
- low performance : the performance is too low for practical use.

Finally, Fig. 2 shows the chosen membership functions for keywords that we associate with the frugality score for the rest of this work. Our output frugality score is chosen to take values within $\Omega_S = [0, 100]$.

After choosing keywords and their relative membership functions for each variable, we define IF-THEN association rules to connect them and create the structure of our frugality score.

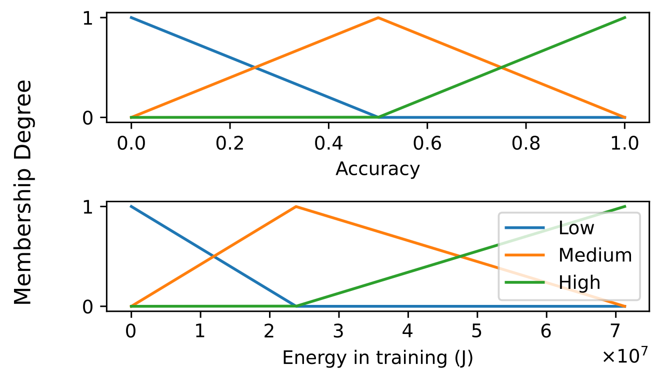


Fig. 1. Examples of membership functions for energy consumptions and accuracy.

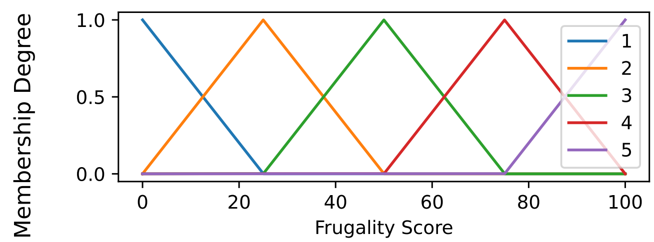


Fig. 2. Membership functions for the output frugality score.

D. Rule definition

To associate each variable, we define rules that associate each keyword configuration of the antecedent variables, the energy consumptions and the performance, to a keyword of the consequent frugality score. For instance, IF the energy consumption is ”low” AND the accuracy is ”high”, the associated score is the highest domain ”5”. We define these fuzzy rules as to ensure the frugality score distribution follows our set of requirements. The chosen rules, shown in Table I, follow the monotonicity requirement of the frugality score respective to the energy and performance, and pulling a ”low” performance or a ”high” energy consumption is equally deficient for the frugality score.

For each rule, the inference on the consequent fuzzy set is modelled by the truncation of the aggregated consequent by the level of truth of the antecedent, so by applying the minimum function on each value of the consequent function with the level of truth of the antecedent. The outputs of each rule are then aggregated into a unique fuzzy set with the OR operator (here the maximum of the membership degree). The output aggregation of the association rules is then translated into a single scalar, here a value for the frugality score, after defuzzification.

E. Defuzzification

Several defuzzification functions Φ are commonly used to interpret the output score fuzzy set as one scalar value [39]

TABLE I
RULE DEFINITION FOR THE FUZZY FRUGALITY SCORE FROM THE ACCURACY AND THE ENERGY CONSUMPTION.

Energy consumption domain D_β	**Accuracy domain** D_α **Low**	**Medium**	**High**
Low	3	4	5†
Medium	2	3	4
High	1	2	3

† Values in the contingency table correspond to frugality score domains D_S.

including the centroid, bisector, Mean of Maxima (MOM), Start of Maxima (SOM) or Last of Maxima (LOM). The centroid and bisector defuzzification functions Φ_{centroid} and Φ_{bisector} are shown in the following equation (5):

$$\Phi_{\text{centroid}} = \frac{\int_{x_-}^{x_+} xM(x)\,dx}{\int_{x_-}^{x_+} M(x)\,dx}$$
$$\Phi_{\text{bisector}} : \int_{x_-}^{\Phi_{\text{bisector}}} xM(x)\,dx = \int_{\Phi_{\text{bisector}}}^{x_+} xM(x)\,dx$$
with $M = \Phi(Max(\Sigma))$ the aggregated Mamdani output, and $x_- = min\{x|x \in \Omega_S\}$ and $x_+ = max\{x|x \in \Omega_S\}$. (5)

These functions are area based, as they respectively calculate the points in the output universe of discourse corresponding to gravity center of the output aggregation, and the point where 50% of the area is located below and above the point. On the other hand, the SOM, LOM and MOM defuzzification functions are not area based. Their values are based on the set of points in the universe of discourse where the maximum membership value is reached in the output aggregation, which is named the *core* of the aggregation. They are shown in equation (6):

$$core(M) = \{\arg\max_x(M(x))\},$$
$$\Phi_{SOM}(M) = min(core(M)),$$
$$\Phi_{LOM}(M) = max(core(M)),$$
$$\Phi_{MOM}(M) = \frac{\int_{core(M)} x\,dx}{\int_{core(M)} dx},$$
with $M = \Phi(Max(\Sigma))$ the aggregated Mamdani output. (6)

Each defuzzification function has a different distribution within the space defined by the input variables. Fig. 3a shows this distribution on our example case. Defuzzification using the centroid and bisector methods induce output distributions that are continuous, while MOM, SOM and LOM show distributions in stages along the input variables. The centroid and MOM functions are commonly used as they are monotonic. The distribution histograms of each score, in Fig. 3b, shows the continuity of imposed by the centroid and bisector defuzzification functions, and the Gaussian profile of the centroid function. As frugality aims for the evaluation of the accordance to a set objective, the centroid method was selected as it provides a continuous and approximately Gaussian distribution, improving interpretability.

The scoring method we produced using this system was implemented into an open access Python package frugalityscore[3] available on GitHub and within the PyPI library.

IV. Methods & Experimental Setup

To evaluate our frugality scoring strategy, we chose to introduce its applications in 2 different common practices in the machine learning community. Each of these execution scenarios presented below carry their own objectives in terms of performance and energy consumption goals, which provides a wide evaluation of the plasticity of the fuzzy based scoring strategy.

A. Experimental scenarios

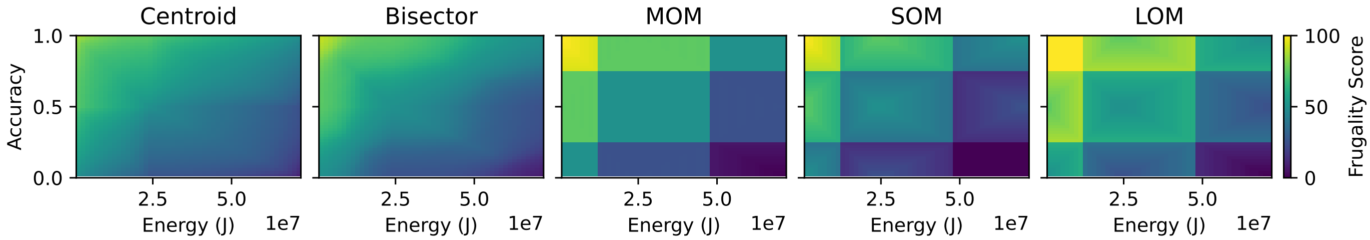
In this work, we identified 2 scenarios of applications of the fuzzy-logic-based frugality evaluation method we presented in section II-B.

The first scenario addresses applications of machine learning models without neural network architectures. The designated task to represent this scenario was classification on the MNIST dataset, with comparisons between classical classifiers available within the package scikit-learn : Nearest Neighbors, Linear SVM, QDA, Naive Bayes, Decision Tree, Random Forest and AdaBoost. Each execution was repeated 30 times for the statistical significance of the energy consumption evaluation. The second scenario focused on the energy consumption of neural network models, with two different applications. We selected 6 benchmark CNN to train and to evaluate their energy consumption for their popularity in the AI community or their relatively low number of parameters : ResNet18 (11M parameters), VGG16 with and without batch normalization (138M), MobileNet V2 (4M), EfficientNet B0 (5M), ShuffleNet V2 (1M), and SqueezeNet V1 (1M). We trained these models on the ImageNet dataset [40] and the CIFAR100 dataset [41]. Each model was trained once from scratch on ImageNet due the high running time and 10 times from scratch on CIFAR100. For each run, we used the cross-entropy loss with the Adam optimizer and a learning rate reduction strategy on metric plateau.

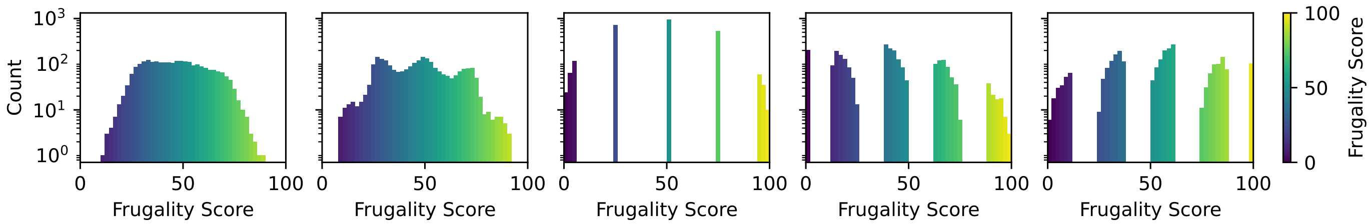
For further proof of our frugality evaluation strategy on a real life application, we applied our fuzzy frugality score on a pre-existing energy oriented dataset provided by Tripp et al. called BUTTER-E [42]. This dataset offers energy consumption estimations of 30582 multilayer perceptron (MLP) configurations, presenting 8 network shapes and tested on 13 datasets. We apply our frugality evaluation strategy on this dataset for its relative exhaustiveness over MLP architectures, and base our evaluation on the standardised energy evaluation given by the authors and the median maximum accuracy reached during training as a performance metric.

The evaluation of the frugality of these applications within these scenarios relies on both identified performance metrics and energy consumption measurements.

[3]https://anonymous.4open.science/r/frugscore/



(a) Frugality score distribution along energy consumptions and accuracy.



(b) Frugality score distribution histograms.

Fig. 3. Example of fuzzy frugality score distribution following fuzzy inference with different defuzzification functions.

B. Data Collection Pipeline

We decided to apply two different strategies, shown in Fig. 4, in accordance with the different applications in our scenarios. The energy consumption in each application was measured using a connected plug Aeotec® Smart Switch 7 connected to a database INFLUXDB through a Z-Wave protocol with Z-WAVE JS and MOSQUITTO, recreating a data collection pipeline from [14]. The energy consumption of the classification on ImageNet, requiring more intensive hardware and thus not being connected to the plug, was approximated using the CODECARBON package based on the CPU, GPU and RAM usage during runtime. To focus on the energy consumptions of the task of interest, we also applied a filtering process of the energy measurements by removing a background energy consumption measured before the experiments.

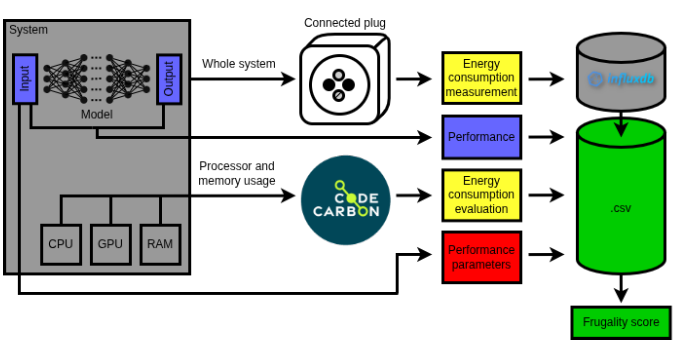


Fig. 4. Data collection pipeline, with a **measurement** strategy using the connected plug and an **evaluation** strategy using CODECARBON for the energy consumption of the system. The data is aggregated into a single CSV file used for the frugality score computation.

C. Evaluation Protocol

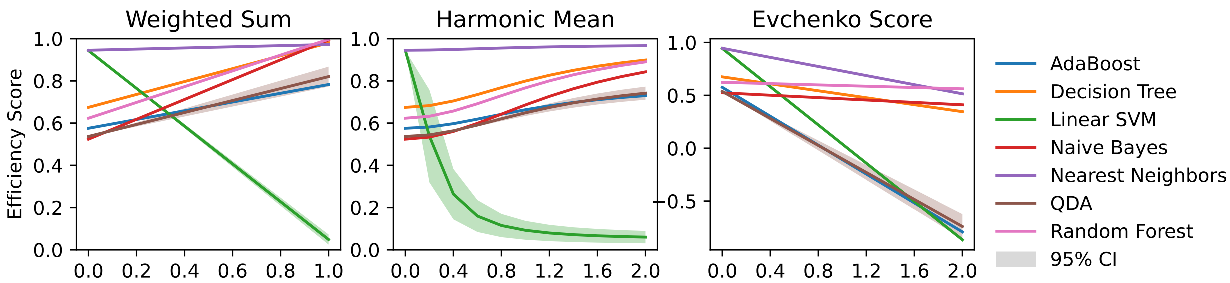
Depending on the scenario, different performance metrics are used to describe the results of the compared methods. In the non-neural based machine learning scenario, this performance was evaluated using the classification accuracy on a randomly selected test sample from the original data, and we used the top-1 classification accuracy in the neural network based machine learning scenario. These choices were motivated by the simplicity of their calculation and their high interpretability, as well as their general practice in the ML community.

Each execution scenario was implemented on an Intel i5-12600 3.30GHz CPU, a NVIDIA RTX™4500 Ada GPU with 24 GB and 2 × 32 GB RAM. All experiment were conducted on Python 3.9, PYTORCH 2.6, SCIKIT-LEARN 1.4.1 and CODECARBON 2.8.3. The energy consumption measurement pipeline was implemented using INFLUXDB, Z-WAVE JS and MOSQUITTO as presented in section IV-B.

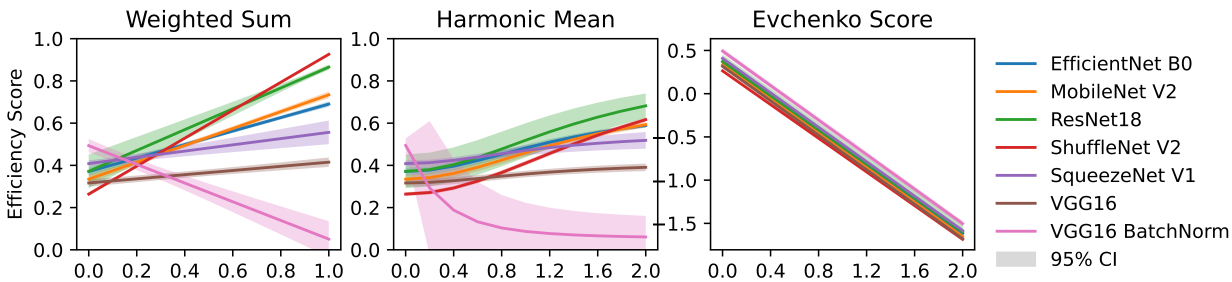
This paper focuses on the energy consumption of the system during the execution phases after hyper-parameter tuning. Therefore, the energy consumed during the hyper-parameter selection for each scenario is not taken into account in our frugality scoring method, but only the energy consumption during training with the selected hyper-parameters, and the energy consumption in inference. The batch size was selected incrementally in order to maximize the GPU usage without over using CPU. Other hyper-parameters were selected using grid search over a few training epochs based on accuracy on the validation split. Overall, this phase required a high computation time and could be taken into account for a more complete frugality evaluation.

V. RESULTS

In this section, we present the application of our fuzzy-logic-based frugality scoring strategy on the different execution scenarios presented in section IV-A, followed by an evaluation of the calculation of energy consumption metrics.



(a) Non-fuzzy efficiency score comparison for non-neural based models on MNIST.



(b) Non-fuzzy efficiency score comparison for deep learning models on CIFAR100.

Fig. 5. Non-fuzzy efficiency score comparison across machine learning scenarios. Filled areas correspond to calculated 95% confidence intervals (CI). X-axes define respective weights given to total energy consumption (training and inference added).

A. *Scoring Method Comparison*

The different efficiency scores available in the literature, presented in section I-B, provide a first insight on the trade-off between the energy consumption and the performance in each scenario. For the classification task on MNIST with non-neural classifiers, Fig.5a shows that the Nearest Neighbors and Linear SVM methods yielded the highest accuracies around 95%, while other methods showed accuracies between 50 and 70%. The Naive Bayes method showed the lowest accuracy, but also the lowest total energy consumption and LinearSVM is the most power-hungry method in this experiment. Based on these measurements, these scores show the evolution of the rank of each method depending on the importance of the energy consumption. The weighted sum score and the harmonic mean score show similar rankings, and isolate the Nearest Neighbors classifier as the most efficient. The Evchenko score, on the other hand, tends to isolate the Random Forest method the more the energy consumption is considered. The application of these scores on the neural-network based classification on CIFAR100 shows very different energy consumption and accuracy distributions, respectively significantly higher and lower. Here, ShuffleNet V1 shows the poorest test accuracy around 26%, while VGG16 BatchNorm shows the highest around 49%, followed by SqueezeNet V1, ResNet18 and EfficientNet B0 close to 40%.. In terms of energy consumption, the lowest consumptions where measured for ShuffleNet V2 and ResNet18. These results translate within the efficiency scores by isolating ResNet18 as the most efficient according to the weighted sum and harmonic mean scores. Yet, The ranking given by these scores differ for the other classifiers, the weighted sum score favorising ShuffleNet V1 in second, and the harmonic mean score ranking it after EfficientNet B0 or MobileNet V2.. The Evchenko score shows a similar ranking than the harmonic mean score, but it relies solely on the accuracy as the energy consumption does not modify it when considered.

Overall, these efficiency scores permit a visualization of the performance and energy consumption trade-offs of each method, but remain difficult to interpret. Their value rely on the weight parameter given to the energy consumption dimension. Therefore, the user must choose a weight that has little to no real-life interpretation. Otherwise, the user can observe the distribution according to a range of weight values, but the ranking given by the different scores varies significantly. An interpretable score based on real-life parameters appears relevant to overcome these issues.

B. *Scoring Across Scenarios*

1) Non-neural Machine Learning: In this scenario, we included energy consumption thresholds for both training and testing based on estimated runtime. Thus, we considered runtimes of 5 seconds, 1 minute and 3 minutes for training, and 1, 5 and 30 seconds for testing. These thresholds translate to energy consumptions with a power estimation of 275 W, and we converted them into triangular membership functions presented in Fig. 6. Based on the literature [43], a higher runtime was expected for the Nearest Neighbor method for training and testing, with a higher accuracy than the Linear SVM and Decision Tree boosting methods.

Our results in Fig. 7 show that our implementation allowed Linear SVM and Nearest Neighbor to yield the highest mean accuracies, with other methods between 50 and 60%. However, Linear SVM showed significantly higher runtime and energy consumptions in training and inference, around 27 kJ and 6 kJ respectively. In this case, our frugality score did not select a specific method with a 100 score because

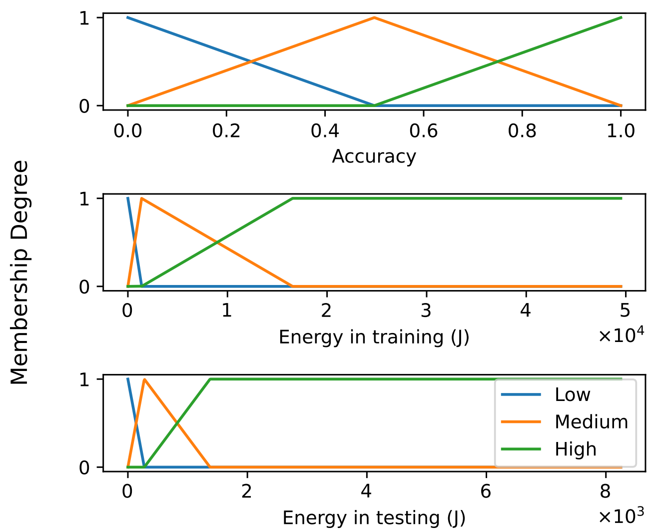


Fig. 6. Membership functions for energy consumptions during training and testing, accuracy and the frugality score for non-neural based models on MNIST.

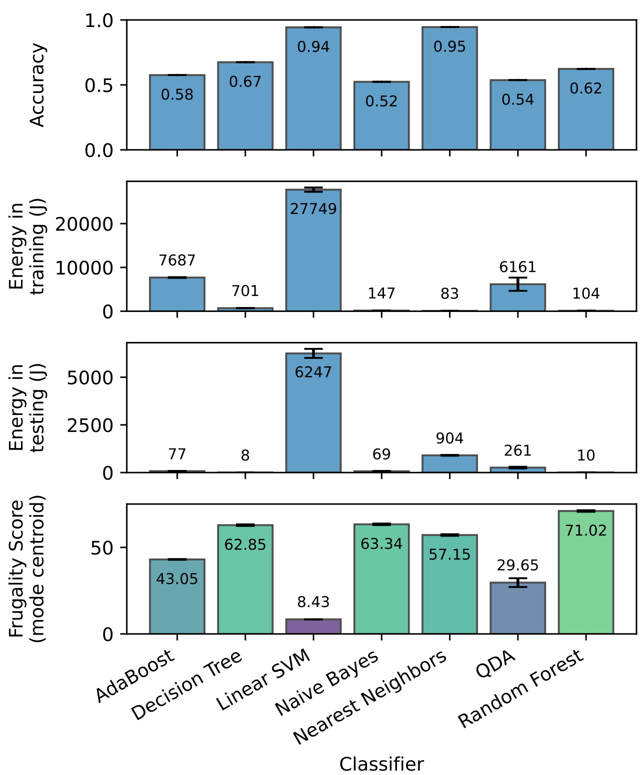


Fig. 7. Mean accuracy, energy consumption evaluated during training and testing and fuzzy frugality score for non-neural based models on MNIST, with centroid defuzzification. Error bars show 95%-confidence interval based on 30 repetitions.

each method showed either medium performances or high energy consumptions. However, the definition of our objectives identified the Random Forest method as the most frugal for its low energy consumption in training and inference despite its medium performance, with total frugality score of 71.02. It also favoured methods with lower energy consumptions, and gave a similar score around 60 to Decision Tree, Naive Bayes and Nearest Neighbors. If the Nearest Neighbors method is the most efficient in terms of performance for energy consumed during training, our preset objectives for energy consumption in inference. Thus, our score favors the choice of Random Forest for this classification tasks, by taking into account the performance and energy consumptions in both training and inference, and provides a interpretable justification of this choice by pointing to the users preset objectives. This frugality approach is relevant as a holistic evaluation of the results of a method, and is particularly relevant for high energy consuming machine learning tasks, like deep learning.

2) Deep Learning: The application of our scoring method in deep learning induced the definition of higher thresholds for the membership functions. First, for the application of this scenario on CIFAR100, we considered running times of 5 minutes and 1 hour to 5 hours in training, and 10 seconds and 30 seconds to 1 minute in testing, based on our own experience with the hardware used. These values translates to respectively 83 kJ, 990 kJ, 4950 kJ and 2.75 kJ, 8.3 kJ, 16.5 kJ for a 275 W system. The associated membership functions are shown in Fig. 8.

Each classifier yielded a lower top-1 accuracy than expected according to the literature [44] under 50 %. Fig. 9 shows that VGG16 with batch normalization yielded higher accuracies but higher energy consumption both in training and inference, which can be explained by the higher number of trainable parameters of VGG16 and VGG16 with batch normalization compared to the other candidate models. These results identify VGG16 without batch normalization as the least frugal model, showing poor top-1 accuracy and relatively high energy consumptions in training and inference, and ResNet18 received a higher score around 40.53. Based on the direct energy consumption and accuracy values, one would struggle to formalize the choice of ResNet18 for its frugality, as its performance is not amongst the highest within our results, but our score favours this model through the interpretable preset objectives.

Training the CNNs on ImageNet was expected to require longer runtimes than CIFAR100 due to the size of the training datasets (respectively 30 k and 1.2 M images). Thus, the training and testing on ImageNet called for higher energy consumption thresholds. We considered running times of 1 day and 3 days to 3 weeks in training, and 10 seconds and 30 seconds to 2 minutes in testing. These thresholds correspond to energy consumptions around $< 2.38 \times 10^4$ kJ, 7.13×10^4 kJ, 4.99×10^5 kJ and 2.75 kJ, 8.25 kJ, 33 kJ respectively for a 275 W system, and the associated membership functions are shown in Fig. 10.

In this application, we obtained a similar top-1 accuracy than the literature [44]. Fig. 11 shows ShuffleNet V2 and SqueezeNet V1 yielded lower accuracies, around 50 %, while other classifiers show similar accuracies around 70 %. However, VGG16 shows the highest energy consumptions in both training and testing, followed by EfficientNet B0 with energy consumptions more than twice lower in training.

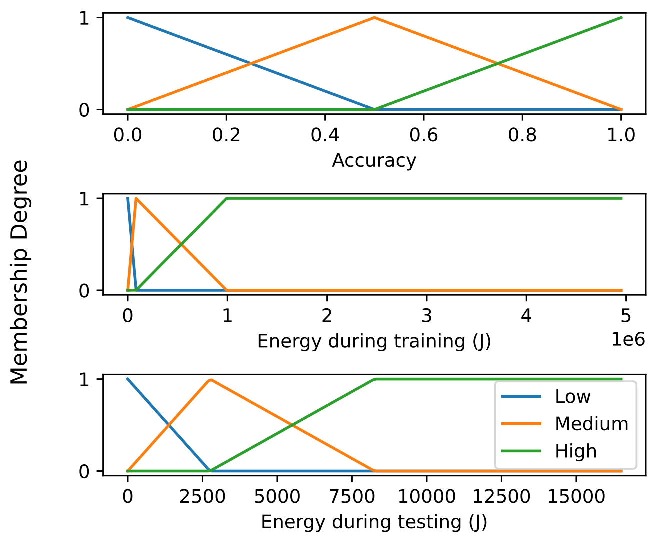


Fig. 8. Membership functions for energy consumptions during training and testing, accuracy and the frugality score for deep learning models on CIFAR100.

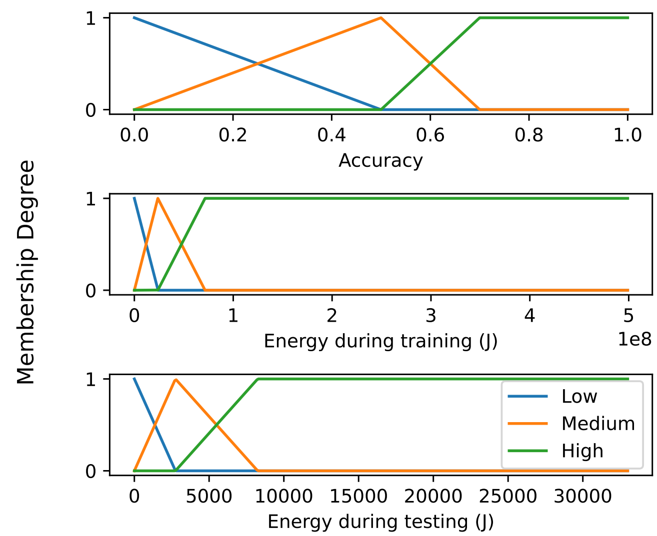


Fig. 10. Membership functions for energy consumptions during training and testing, accuracy and the frugality score for deep learning models on ImageNet.

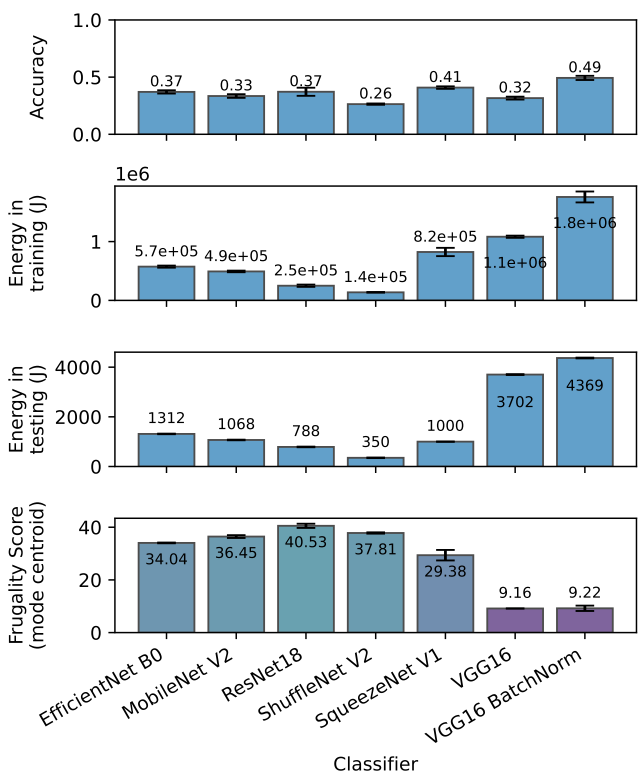


Fig. 9. Mean top-1 test accuracy, energy consumption evaluated during training and testing and fuzzy frugality score for deep learning models on CIFAR100, with centroid defuzzification. Error bars show 95%-confidence interval based on 10 repetitions.

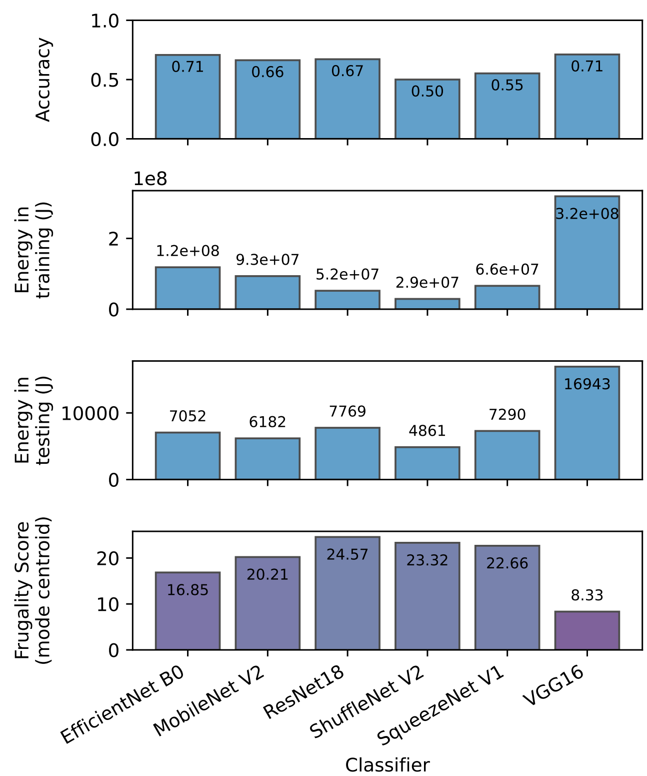


Fig. 11. Top-1 test accuracy, energy consumption evaluated during training and testing and fuzzy frugality score for deep learning models on ImageNet, with centroid defuzzification.

Here, energy consumptions in training fell under the keywords ”medium” to ”high” for all classifiers both in training and testing, which lead to low frugality scores with values under 25. Again, in this case, no classifier truly appears frugal based on our preset objectives. But a relative comparison of these classifiers identified ResNet18 as the most frugal from our benchmark models still, with a score around 24.57. Indeed, this classifier yielded a medium performance but a lower energy consumption than others in both training and testing.

Thus, this classifier is preferred according to our objectives.

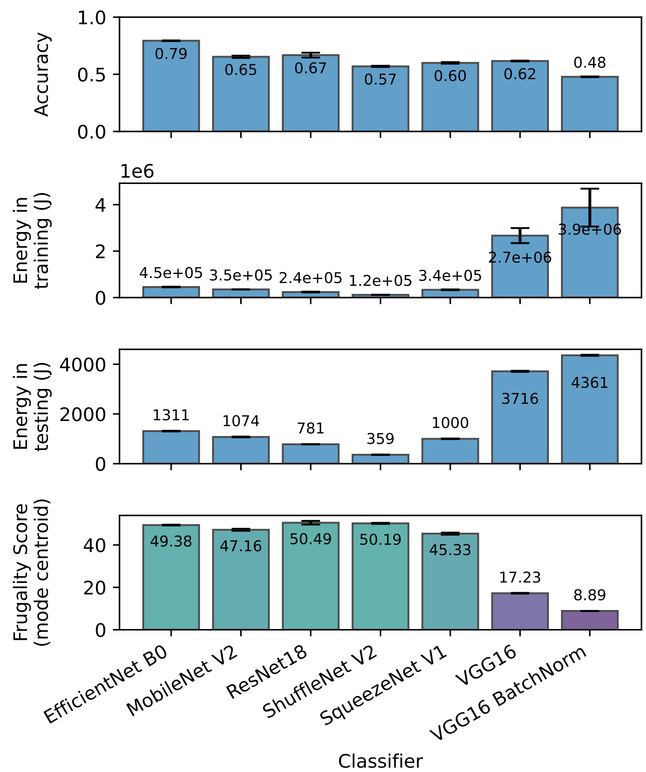


Fig. 12. Mean top-1 test accuracy, energy consumption evaluated during training and testing and fuzzy frugality score for deep learning models on CIFAR100 pretrained on ImageNet, with centroid defuzzification. Error bars show 95%-confidence interval based on 10 repetitions.

As transfer learning is a common practice in machine learning to reduce the training time of a deep learning model while adapting its weights to the target dataset, we tested our frugality scoring method on the classification task of CIFAR100 using models pre-trained on ImageNet. Here, the energy consumptions considered correspond to the fine-tuning and inference on CIFAR100, all tested using the same classifications models as in the experiment with the training on CIFAR100 from scratch. The expected results were higher top-1 accuracies than with training from scratch, a similar energy consumption in testing, and a lower energy consumption during training for each model considering an early stopping strategy occurring after fewer training epochs, except for VGG16 and VGG16 BatchNorm for their higher number of parameters.

Fig. 12 shows that each model yielded a lower energy consumption in training and a higher accuracy than when trained from scratch except VGG16 BatchNorm which showed similar accuracy for a higher energy use. Using the same membership functions as for the trainings from scratch on CIFAR100, the calculated frugality scores are thus similar for these models as when trained from scratch in terms of ranking. On the other hand, the absolute value of the frugality scores for each better performing model showed higher frugality evaluations with the transfer learning approach. As the accuracy of each model increased with transfer learning, so its frugality score. ResNet18 still yields higher frugality scores due to its higher accuracy, and the decrease of the energy consumption during training and inference.

After testing our scoring approach on our own experimental results for common scenarios, it is essential to validate this approach on benchmark energy consumption datasets.

C. Validation on BUTTER-E

As a proof of concept, we applied our framework to the BUTTER-E dataset. BUTTER-E [42] is a benchmark dataset based on BUTTER [45] which presents the training results of around 3.55 million individual runs of multi-layer perceptron (MLP) architectures on different classification datasets. In total, this dataset shows the results of 8 network shapes (overall network structure), 14 depths and 20 sizes (number of trainable parameters), trained on 13 datasets and on 6 hardware configurations.

We selected the runs of rectangular shaped MLP with 2,048 trainable parameters, trained on MNIST [46] using two Intel Xeon Gold 6154 CPU and two NVIDIA V100 PCIe GPU to conduct the application of our scoring method, and an overall idle power of the system around 900 W. To define the membership functions used in the frugality score model for this application, we identified energy consumptions thresholds around 3.24 MJ, 9.72 MJ and 19.4 MJ corresponding to idle consumptions of the system for 1, 3 and 6 hours. These thresholds were used for the definition of the energy consumption membership functions, shown in Fig. 13.

The energy consumptions during training and median accuracies relative to the depth of the model are shown in TABLE II, along the calculated frugality scores. Each model in this experiment show a similar structure, with the same number of trainable parameters, but with a different number of layers. Their training on MNIST show that the model accuracy decreases as the number of layers increases, and we observe the opposite effect on the energy consumption during training. Following these trends, the calculated frugality scores also decreases when the number of layers increases. This experiment provides a clear example of the ideal case in terms of both frugality and efficiency. The 3-layer classifier is the most efficient here, with the highest accuracy and the lowest energy consumption. It is also the most frugal among the studied models, but just more than average in relation to the preset objectives.

VI. Discussion

Efficiency and frugality are two different concepts that encapsulate different strategies in terms of sustainability evaluation. They both aim for an increase of the performance while reducing the use of costly resources. In this aspect, our worked showed that both approaches result in similar evaluations for certain applications. Both approaches prefer low-depth MLP in BUTTER-E. Yet, when the number of variables to consider is higher, efficiency remains hard to quantify while retaining interpretability. Frugality mitigates these issues, and provide real-life interpretable results.

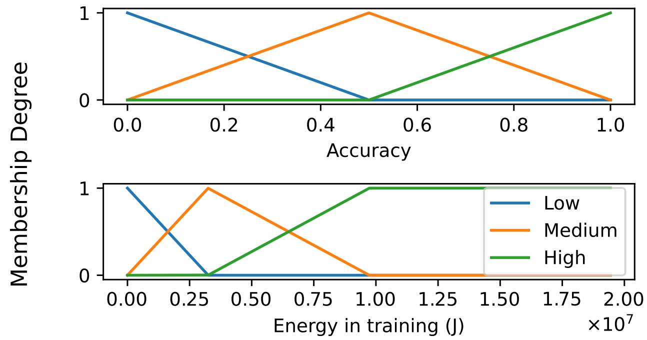


Fig. 13. Membership functions for energy consumptions during training and accuracy for MLP on MNIST from BUTTER-E.

TABLE II
ACCURACY, ENERGY CONSUMPTION EVALUATED DURING TRAINING ON NVIDIA V100 PCIE AND FUZZY FRUGALITY SCORE FOR 2048-PARAMETER RECTANGLE MLP ON MNIST FROM BUTTER-E.

Model depth	Median Accuracy†	Energy Consumption (MJ)	Frugality Score‡
3	**0.80**	**2.0**	**66.35**
4	0.78	2.1	65.25
5	0.74	2.1	63.82
6	0.64	2.1	60.58
14	0.11	2.4	40.88
18	0.11	2.5	39.89
20	0.11	2.7	38.37

† Median of maximum accuracy on test set reached during training.
‡ Defuzzified score carried out by centroid calculation.

As this paper presented results from several experimental setups with high runtime processes, a higher number of experiment runs would benefit a statistical significance of each measurement. As Green AI is a growing field in today's research, few datasets including energy consumptions measurements are available. The application of our strategy to BUTTER-E aimed to mitigate this aspect, but an extension of this application to other datasets is to be expected.

The energy consumption measurements also suffered from the precision of the sensor. The connected plug used for empirical measures allowed a minimal measurement period of 30 seconds, and an error rate of 3 W for the power measurement. Our study also shown how CODECARBON tends to underestimate energy consumptions.

This study also disregarded energy consumptions related to implementation tests and hyper-parameter tuning. In this sense, the global energy consumption of a method was only defined by its direct usage, without the development stage.

Finally, only energy consumptions were taken into account in this study, while recent research show a significant impact on water consumption, electronic waste and human equity [47]. Applying a similar strategy on these aspects is necessary to further deepen the precision of a frugality evaluation.

VII. CONCLUSION

To avoid falling in the domain efficiency and focusing on reducing the energy consumption of a system while improving its performance, frugality questions the quality of performance and energy consumption measures related to preset objectives. This work presented different common scenarios in computer science for which high performance and low energy consumptions have different meanings. In applications outside of machine learning, a single energy consumption can be relevant to estimate the ecological cost of a method. However, sustainability evaluation for machine learning applications need to include energy consumptions both in training and testing, or inference in deep learning. Our fuzzy logic based scoring method captures this concept by using objectives predefined by users to identify high energy consumptions and sufficient performances specific to the case study. We showed that this scoring strategy permits a frugality assessment of a method, with low scores given to extensively large deep learning classifiers with poor performance for their energy consumptions, and higher scores to smaller models with a similar or slightly lower performance, allowing a relative analysis for off-the-shelf model selections.

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APPENDIX

TABLE III
PARAMETERS USED FOR MEMBERSHIP FUNCTION DEFINITIONS FOR EACH FRUGALITY SCORE CALCULATION.

Scenario	**Dimension**	**Modality**	**a**	**b**	**c**	**d**
Theoretical example	Accuracy	Low	0	0	0.5	0.8
		Medium	0.5	0.8	0.8	1
		High	0.5	0.8	1	1
	Energy in training (J)	Low	0	0	0	23760000
		Medium	0	23760000	23760000	71280000
		High	23760000	71280000	71280000	71280000
Non-neural machine learning on MNIST	Accuracy	Low	0	0	0.5	0.8
		Medium	0.5	0.8	0.8	1
		High	0.5	0.8	1	1
	Energy in training (J)	Low	0	0	0	275
		Medium	0	275	275	1375
		High	210	1375	49500	49500
	Energy in testing (J)	Low	0	0	0	137.5
		Medium	0	137.5	137.5	275
		High	137.5	275	16500	16500
Deep learning on ImageNet	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	0.75
		High	0.5	0.75	1	1
	Energy in training (J)	Low	0	0	0	23760000
		Medium	0	23760000	23760000	71280000
		High	23760000	71280000	498960000	498960000
	Energy in testing (J)	Low	0	0	0	2750
		Medium	0	2750	2750	8250
		High	2750	8250	33000	33000
Deep learning on CIFAR100	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	1
		High	0.5	1	1	1
	Energy in training (J)	Low	0	0	0	82500
		Medium	0	82500	82500	990000
		High	82500	990000	4950000	4950000
	Energy in testing (J)	Low	0	0	0	8250
		Medium	0	8250	8250	16500
		High	8250	16500	33000	33000
Deep learning on MNIST from BUTTER-E	Accuracy	Low	0	0	0	0.5
		Medium	0	0.5	0.5	0.75
		High	0.5	0.75	1	1
	Energy in training (J)	Low	0	0	0	3240000
		Medium	0	3240000	3240000	9720000
		High	3240000	9720000	19440000	19440000